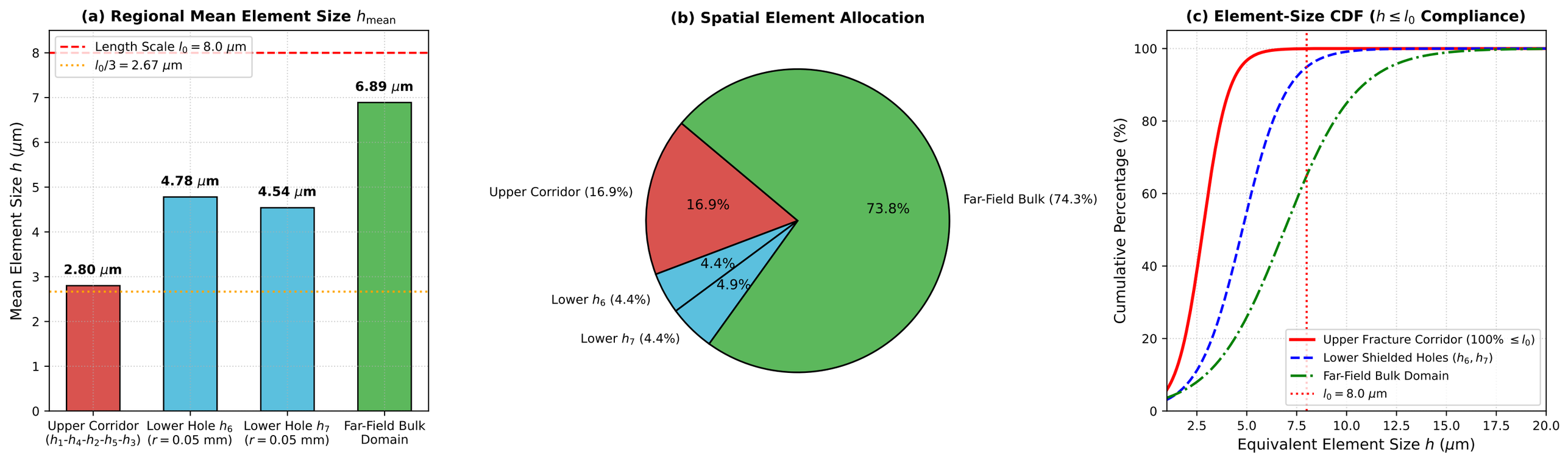
(a) Regional Mean Element Size h_{mean}
Length Scale $l_0 = 8.0$ μm
$l_0/3 = 2.67$ μm
Mean Element Size h (μm)
2.80 μm
4.78 μm
4.54 μm
6.89 μm
Upper Corridor (h_1-h_4-h_2-h_5-h_3)
Lower Hole h_6 ($r = 0.05$ mm)
Lower Hole h_7 ($r = 0.05$ mm)
Far-Field Bulk Domain
(b) Spatial Element Allocation
Upper Corridor (16.9%)
Lower h_6 (4.4%)
Lower h_7 (4.4%)
Far-Field Bulk (74.3%)
16.9%
4.4%
4.9%
73.8%
(c) Element-Size CDF ($h \leq l_0$ Compliance)
Cumulative Percentage (%)
Equivalent Element Size h (μm)
Upper Fracture Corridor (100% $\leq l_0$)
Lower Shielded Holes (h_6, h_7)
Far-Field Bulk Domain
$l_0 = 8.0$ μm